ECONOMIC ASPECTS

Nuclear Power and Energy Security

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The first duty of government is to keep its citizens safe and country secure, and the provision of uninterrupted energy services is increasingly integral to achieving those aims. Energy security represents a public good that markets are generally unable to provide for at an appropriate level. As such, security of energy supply is a principal concern for – and the responsibility of – all governments.

How energy security is defined and pursued through policy will depend on the situation of an individual or a country, and the timeframe considered. Following the Covid-19 pandemic and the onset of war between Ukraine and Russia in 2022, governments have become more aware of the vulnerabilities of cross-border supply chains in key sectors. This reassessment of trade-related risks has prompted efforts to increase energy security, and supply chain security in general.

The 1970 energy crises led to a major expansion of nuclear capacity as countries sought to diversify their sources of energy supply and reduce dependence on the continual import of large volumes of fossil fuels. For many countries, the appraisal of recent events and, in particular, high fossil fuel prices, has led to similar decisions that may result in a greater role for nuclear energy.

Historical perspective

Energy security emerged as a concept in the first half of the 20th century as political and military leaders showed concern over available fuel supplies for armed forces. Countries sought to ensure supply through various measures during the Second World War, including substituting foreign supplies with domestic resources,

Privacy - Terms

restricting non-essential uses at home, and battling for control over oil fields in Indonesia, the Middle East and elsewhere[1].

After the Second World War the importance of oil continued to grow, underpinning the progress and growth of industrialized economies. Oil became essential for transport, food production, electricity generation, health care and heating in developed nations.

Production was dominated by the companies known as the 'Seven Sisters', which then controlled some 85% of the world's oil. However, in 1960, Saudi Arabia, Iran, Iraq, Venezuela and Kuwait formed the Organization of the Petroleum Exporting Countries (OPEC), and by 1970, OPEC controlled half of the world's oil supply.

In 1973 Arab countries, in protest at the USA's support for Israel, cut supplies of oil. The embargoes in the 1970s made apparent the new power of oil producing nations, and the vulnerability of industrialized countries that increasingly relied on the hydrocarbon but did not produce enough to satisfy their needs. The response to the embargoes was significant. Importing countries began substituting oil in electricity generation with other energy sources (natural gas, uranium and coal) and built strategic reserves. In addition the United States and other western countries took measures that sought to make disruptions of oil flows to industrial countries less likely in the future. These included: the Carter doctrine, which explicitly stated that the US would, if necessary, use force in the Persian Gulf to ensure the "free movement of Middle Eastern oil"[2]; the fostering of a global oil market; and the establishment of the International Energy Agency (IEA), the founding mission of which was to coordinate responses to, and reduce the impact of, oil disruptions among OECD countries.

By the turn of the century, concerns about security of energy supplies had extended beyond the geopolitical risk to reliable fossil fuel supplies. Over time energy security policies have broadened to encompass an engineering perspective, recognizing the need to understand and manage the vulnerabilities of increasingly complex systems for energy distribution and electricity generation and distribution.

These separate components to energy security cannot be managed effectively in isolation. For example, a decision to substitute fossil fuels in electricity generation with significant amounts of variable renewable technologies may on the one hand increase 'domestic' production, but on the other will require fundamental changes to the design and operation of electrical grids to ensure their

continued stability and reliability, and will make production less predictable.

As such, organizations such as the IEA now encourage policymakers to take a more comprehensive view of energy security to avoid simply substituting one vulnerability with another.

Box 1: Geopolitics and engineering perspectives

Before the oil crises of the 1970s, energy security was viewed from a geopolitical perspective. The term was used principally by developed countries concerned about the supply of oil and coal at a stable price.

Over time the concept has broadened to include other fuels, as well as the system and infrastructure for delivering available fuels. This engineering perspective has historically been concerned primarily with the reliable operation of energy technologies, such as refineries and power plants. In recent years more attention has also been given to electricity systems. The reasons for this are many and include:

- The growing importance of electricity in transport, domestic heating and industrial processes.
- Increased penetration of variable renewable energy, requiring fundamental change to electricity systems and increased reliance on cross-border electricity transmission and exchanges.
- The erosion of capacity margins in electricity systems to meet peak demands.
- The vulnerability of electricity networks to short-term disruptions.
- Increasing digitalization of energy systems.

Defining energy security

The modern concept of energy security is complex and there is no singular definition of what energy security is. It is generally defined as ensuring that people have access to an uninterrupted supply of the energy that they need, at a price that is affordable to them[3]. However, the 'need' of a country or an individual, and the risks

facing energy systems, differ markedly – and the appropriate balance between secure supplies and cost is subjective.

As a result, different stakeholders have different perspectives on energy security, leading to a diversity of definitions in academic literature. Its meaning is different in different markets and over different timescales, for example, and the concept of national energy security will differ fundamentally based on what is the norm for different countries.

According to the World Health Organization, one third of the world's population still rely on wood, crop residues and animal waste for cooking and heating[4]. For those 2.4 billion people, the definition of energy security clearly differs markedly from that that would be applied in industrialized countries where consumers are accustomed to a reliable and affordable supply free from disruptions. To many, therefore, energy security means energy to supply the most basic of human needs like clean water and sanitation, with insufficient capacity and rapid demand growth likely to predominate as concerns for government. In more developed parts of the world, energy security is more about protection from supply interruptions and price volatility, with import dependency and aging infrastructure likely to weigh heavy.

Academics have sought to establish classifications for the many different dimensions of energy security. Examples are shown in the table below.

Alhajji[5]	von Hippel et al.[6]	Sovacool & Brown[7]
Economic	Energy supply	Availability
Social	Economic	Accessibility
Environmental	Technological	Affordability
Foreign policy	Environmental	Acceptability
Technical	Social/cultural	-
Security	Military/security	-

Broader energy policy and energy security

Across much of the world, broad government policies typically reflect three dimensions: security, economics and environment. Whilst each perspective goes far beyond energy, energy policy is

critical to each and generally viewed in the same framework. For example:

- Japan: "3Es" – energy security, economic efficiency, environment[8]
- UK: secure, clean and affordable[9]

Energy security therefore cannot be viewed in isolation. The 'Energy Trilemma' concept identifies the need to find a balance between energy security, affordability and sustainability. Whilst not necessarily in conflict, the trilemma concept is an acknowledgment of competing demands, and the fact that for an appropriate balance to be found, the three aspects must be considered together. For example, energy supply cannot be secure if it is not affordable, and so affordability is an important part of most definitions of energy security.

Common terms: sovereignty, robustness and reslience

Three terms are often used by politicians and policy makers to discuss energy security in the 21st century: sovereignty, robustness and resilience. Whilst used interchangeably, Cherp and Jewell[10] have described the distinct aspects of energy security that these terms refer to:

Sovereignty originates from the initial focus on oil supplies during WW2 and the 1970s crises. Focusses on the threats posed by the intentional actions of external malevolent actors, such as embargoes. Risk minimization strategies include reducing import dependency by increasing domestic production (i.e. seeking energy "independence" or "domestic" sources of supply), diversifying suppliers, and pursuing efficiency measures.

Robustness focusses on protection from factors that are relatively predictable. They may be natural (e.g. resource scarcity, climate change), technical (e.g. aging infrastructure) or economic (e.g. rising prices) in nature and are generally long-term.

Resilience focusses on protection from factors that are relatively less predictable (e.g. political instability, extreme weather, pandemics) and generally short-term.

The contribution of nuclear energy

Nuclear energy has characteristics that can strengthen a country's supply security when part of its energy mix:

- Reliable, long-term electricity supply – nuclear power plants provide large amounts of electricity predictably, with reactors achieving high average capacity factors. Plants built today will operate for 60 or more years. As the electricity from nuclear plants is reliable, nuclear can directly displace fossil fuels from a country's electricity mix, reducing a country's import dependency and diversifying its energy mix.

- Energy density – Uranium is a uniquely concentrated source of energy which is easily and cheaply transportable. The quantities needed are very much less than for coal or oil; 30 tonnes of fabricated nuclear fuel are required per year for a 1GW power plant compared with over 3 million tonnes of coal. It is therefore intrinsically a very portable and tradeable commodity. Critically, this means that it is easy to establish strategic inventories; fuel supply for up to two years is typically stored at power plants.

- Geographical and political availability – Uranium is relatively abundant and fuel supplies are spread among politically diverse countries (see below), reducing the risk of supply disruptions.

- Predictable costs – fuel costs for nuclear plants are a minor proportion of total generating costs. The cost of electricity supplied from nuclear plants is therefore predictable over the course of a plant's operational lifetime.

- Resilient infrastructure – nuclear plants are built to withstand and continue operating in extreme weather, as proven during hurricanes and freezing temperatures in the United States and other places.

- Ancillary grid services – nuclear plants can contribute substantially to the provision of ancillary services such as flexible operation to aid frequency control and provision of grid inertia. Nuclear can also help assimilate variable renewable energy without increasing dependency on imported natural gas.

- Heat and power – for most major industrial heat applications, nuclear energy is the only credible non-carbon option. Using nuclear for process heat can reduce demand for oil and natural gas.

The energy crisis that began in 2021 and was greatly exacerbated by Russia's invasion of Ukraine has led to many countries reassessing their plans for nuclear energy:

"*An important part of the British Energy Security Strategy is our plan to deliver eight new nuclear reactors, the equivalent of one a year, powering millions of homes with clean, safe, and reliable nuclear energy.*"
Boris Johnson, former UK Prime Minister – April 2022

"*I will seek measures to reasonably utilize nuclear energy, which is a major means of energy security and carbon neutrality…*"
Yoon Suk-Yeol, President South Korea – May 2022

I believe that nuclear power is important as we work towards carbon neutrality while ensuring energy security."
Yasutoshi Nishimura, Japan's Minister of Economy Trade, and Industry – September 2022

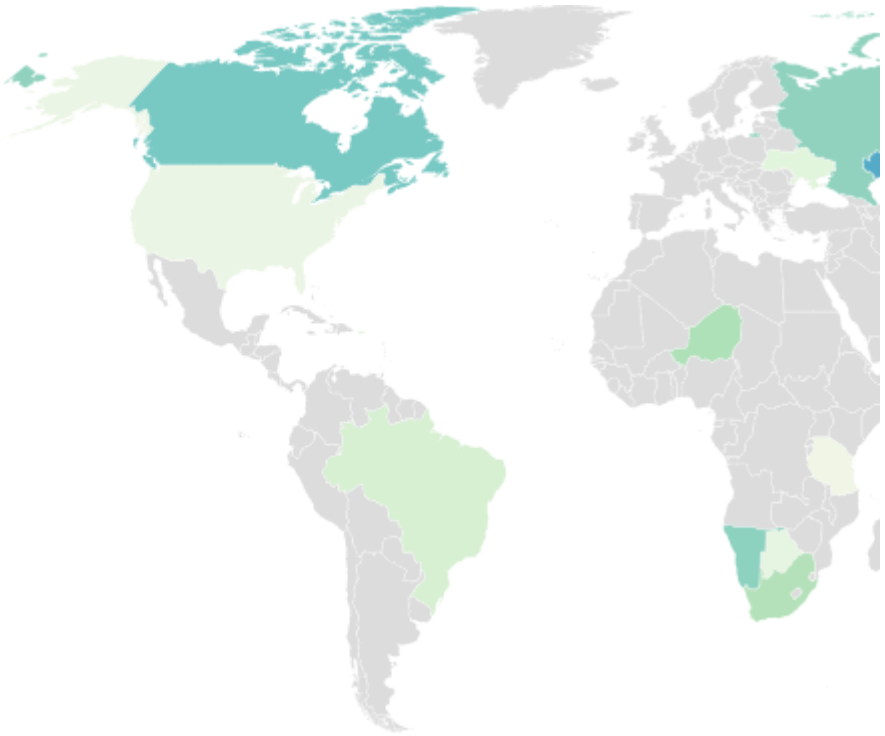
Energy security and the nuclear fuel cycle

To prepare mined uranium for use in a nuclear reactor, several industrial processes are carried out. Initial processing of the product extracted from mines generally takes place close to the mine site. Conventional mines extract the ore which must then be milled to isolate the uranium minerals, whereas in-situ or solution mines leave the ore in the ground, removing the need for this step. After initial processing the product – 'yellowcake' – contains approximately 85% uranium by weight. The relatively small volume of material involved thereafter allows for the flexible siting of subsequent stages of the nuclear fuel cycle; conversion, enrichment and fuel fabrication facilities are operated commercially across Europe, Asia and the Americas.

In response to Russia's military offensive, western countries have sought to diversify their supply of nuclear fuel and fuel services. For more information see Fuel cycle implications section of information page on Russia-Ukraine War and Nuclear Energy.

See also information page on Nuclear Fuel Cycle Overview.

Uranium resources to $130/kg U by country in 2021 (reasonably assured resources plus inferred resources) (source: OECD NEA & IAEA)



International trade, market liberalization and relative risks

It is often asserted that a country can improve its energy security by reducing imports and producing more 'domestic' energy. Whilst the logic is clear, reducing imports does not inevitably increase energy security. Domestic energy supplies may be insecure: Brazil generates two thirds of electricity domestically from hydropower, but droughts over the last decade have resulted in plants operating at a fraction of their capacity with serious socio-economic impacts[11]; in the UK in 1974, miners' strikes led to three-day working weeks[12].

Moreover, the advantages of trade apply to energy imports and exports. The resources available are geographically restricted; they cannot be produced elsewhere through labour or capital expenditure. As a result, most countries that consume significant amounts of energy benefit from imports accounting for significant proportions of their supply; energy costs would otherwise be higher. Conversely, the benefits of international trade for those countries endowed with resources are significant too; the price of oil and gas in international markets, for example, greatly exceeds the cost of extraction.

What is critical, therefore, is consideration of the relative costs and risks of policies. For example, as economies have become more closely integrated through international trade over the past 50 years, many countries have moved towards liberalization of markets for a wide range of goods and services. The movement towards liberalization in electricity markets introduced competition to the procurement and delivery of final energy. This process has generally reduced costs, in part by reducing spare network and generating capacity, and the utilization of 'just-in-time' deliveries. It has, however, reduced reserve margins and system 'buffers', potentially increasing the risks of local shortages to consumers.

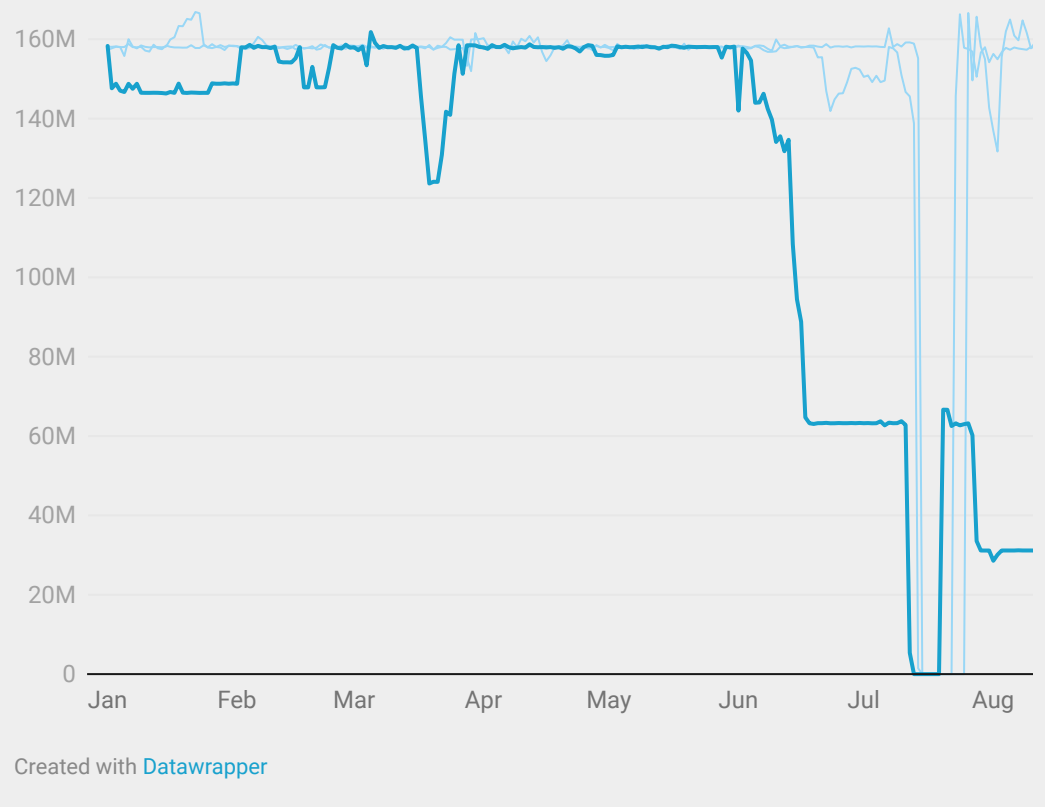
Box 2: Ukraine-Russia war and the weaponization of energy

Russia is the world's largest exporter of fossil fuels, and is a particularly important supplier of gas and coal to Europe. In response to Russia's invasion of Ukraine, European leaders agreed, as part of a broad set of sanctions, to phase out imports of Russian fossil fuels. In response, Russia curtailed supply to Europe, making it difficult for countries to fill gas storage ahead of the winter period.

The actions taken by both sides reflect the political and economic importance of energy, and the different perspectives on energy security for producers and importers. For Russia, like all energy producers, energy security means security of demand – the assurance that its production will be purchased at a fair price over the long term. The proposed ban on imports by Europe recognizes the vital importance of export revenues to Russia's economy. For Europe, like all energy importers, energy security means security of supply. Russia's curtailment of gas supply recognizes the unique exposure of Europe as a major gas-consuming region that is heavily dependent on imports.

For Europe, there is no real substitute in the short term for Russian natural gas. In 2021 European countries imported about 155 bcm of Russian natural gas via pipeline, accounting for about 40% of the continent's needs and equivalent to about half of all liquefied natural gas (LNG) traded by sea each year. Replacing the efficient supply of gas via large-diameter pipelines with LNG imports will incur added costs for Europe. For Russia, there is no short term replacement market for its natural gas given the infrastructure involved in its delivery.

Nord Stream 1 gas flows in cubic metres per day (sources: ENSTOG transparency platform and Nord Stream)



Renewables and energy security

Most mechanisms and strategies for enhancing energy security are concentrated on minimizing the risk of disruptions in hydrocarbon supply, or the occurrence of intolerable price spikes. The widespread deployment of renewable technologies is often touted as a means of reducing a country's vulnerability to such uncertainties.

The main advantage of renewable technologies from a security of supply perspective is that they are based on 'energy flows'. In other words, they do not require the import of fuel. However, the unique characteristics of such technologies presents policymakers with novel and significant geopolitical and engineering energy security challenges to consider.

External factors

A significant expansion of wind and solar power, as well as other technologies associated with a transition from fossil fuels, will create a burgeoning demand for minerals. The lower energy density of intermittent renewable energy compared with fossil fuels and nuclear energy translates directly and inexorably to a greater mineral/material demand per unit of energy. Estimates vary but producing electricity from wind and solar typically increases the quantities of materials requiring extraction, processing and

handling by a factor of at least 10. (For more information see Mineral Requirements for Electricity Generation).

Countries will not rely on continuous access to these goods in the same way as they do today with oil. However, as demand for key minerals grow rapidly, they are likely to be subject to the same price volatility, geopolitical influence and supply disruptions that have long-characterized oil and gas markets. In many ways, the risks of market tightness and price volatility are greater for many of the critical minerals than for oil and gas due to the consolidated nature of supply. For example, the Democratic Republic of Congo produced 70% of the world's cobalt and China 60% of the world's REEs in 2019. Processing operations are more concentrated, with China refining 50-70% of the world's supply of lithium and cobalt, and 90% of its REEs[13].

The IEA's report on The Role of Critical Minerals in Clean Energy Transitions[13] notes that countries need to make sure their energy systems remain resilient and secure as they increase efforts to reduce emissions. "Today's international energy security mechanisms are designed to provide insurance against the risks of disruptions or price spikes in supplies of hydrocarbons, particularly oil. Minerals offer a different and distinct set of challenges, but their rising importance in a decarbonizing energy system requires energy policymakers to expand their horizons and consider potential new vulnerabilities. Concerns about price volatility and security of supply do not disappear in an electrified, renewables-rich energy system"

Engineering factors

A system dependent on weather conditions creates unique vulnerabilities and fundamental challenges for policymakers considering security of supply . Geographical diversification of renewable energy assets is seen as advantageous for helping to manage these risks. As such, deployment of large amounts of variable renewable energy generally results in increased electricity imports and exports via interconnectors between countries. These systems of trade are essential to smoothening variable production and matching supply and demand. Large proportions of variable renewable electricity production therefore result in new interdependencies; they are not likely to result in increased energy 'independence' in the sense often suggested.

Accommodating large amounts of diffuse, variable production technologies puts grid systems under considerable strain. Grid systems that have high levels of wind and solar penetration will need to undergo significant upgrade and expansion to ensure reliability, and will need to be sized such that they can cope with

peak supply from dispersed intermittent sources as well as meeting peak demand during periods when those assets are not producing electricity.

Measuring energy security

There is no accepted quantitative measurement of energy security. Various indicators have been proposed (see table above). The World Energy Council ranks countries' energy security based on:

- Import dependence – the reliance on net imports for total energy consumption and diversity of suppliers
- Diversity of electricity generation – diversity of domestic electricity generation sources
- Energy storage – ability to meet demand for oil and gas considering infrastructure capabilities including storage and refining capacity

Top-10 ranking countries for energy security in 2023 (source: World Energy Council)

In its 2021 Index, the World Energy Council ranked Germany as having the 10th best energy security globally, demonstrating the difficulty in assessing and quantifying energy security risks. Germany has found itself uniquely exposed to the disruptions in energy flows between Russia and Europe that have arisen following the onset of war between Russia and Ukraine (see Box 2).

Prior to the war, over half of the gas consumed in Germany was delivered by pipeline from Russia.

Notes & references

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Appendices

Nuclear Energy and Sustainable Development

Related information

Mineral Requirements for Electricity Generation